MULTIPLEXER

COMBINATIONAL LOGIC CIRCUITS

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TOPIC OUTLINE

Synthesis of Logic Functions

Shannon’s Expansion Theorem

SYNTHESIS OF LOGIC FUNCTIONS

MULTIPLEXER

A **multiplexer** circuit has several data inputs, one or more select inputs, and **one output**. It passes the signal value on one of the data inputs to the output.

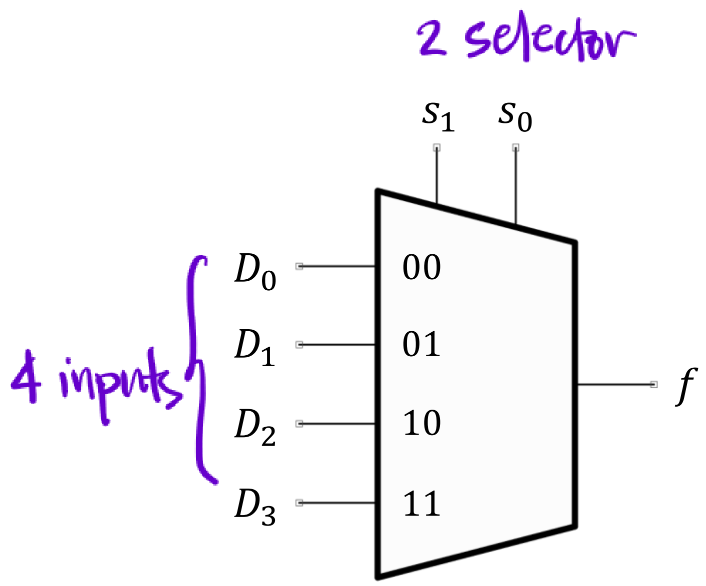
A multiplexer that has n data inputs ($D_0, D_1, \ldots D_{n-1}$), requires $\lceil \log_2 n \rceil$ select inputs.

$\log_2 (n = 2^s)$

$s = \log_2 n$ no. of inputs

no. of selector

4-to-1 Multiplexer



$n = 2^s$ $\quad s = \log_2 n$

$n = 2^2$ $\quad s = \log_2 4$

$n = 4$ $\quad s = 2$

2-TO-1 MUX

Graphical Symbol

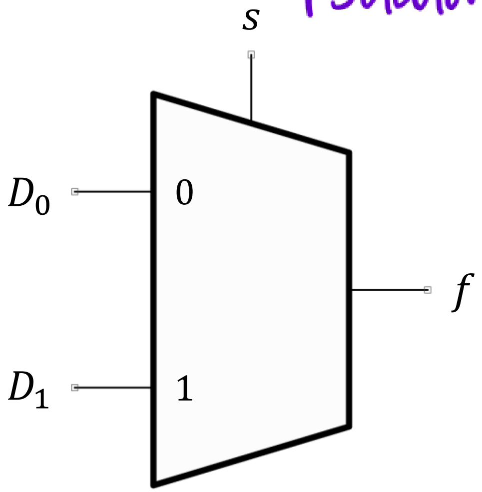
$n = 2^s$

$n = 2^1$

$n = 2$

2 inputs

1 selector



Truth Table

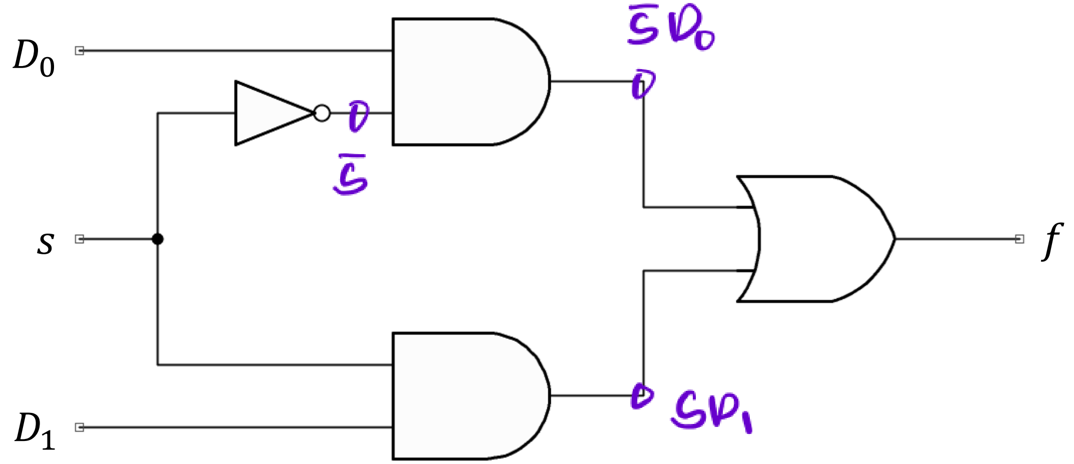
s	f
0	D_0
1	D_1

Minterm

$\bar{s}D_0$

sD_1

Sum-of-Products Implementation



$f = \bar{s}D_0 + sD_1$

4-TO-1 MUX

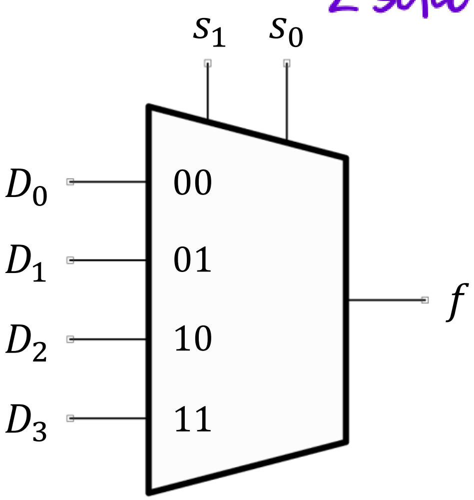
Graphical Symbol

2 selector

$n = 2^s$

$n = 2^2$

$\underline{n = 4}$



Truth Table

s_1	s_0	f
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

Minterm

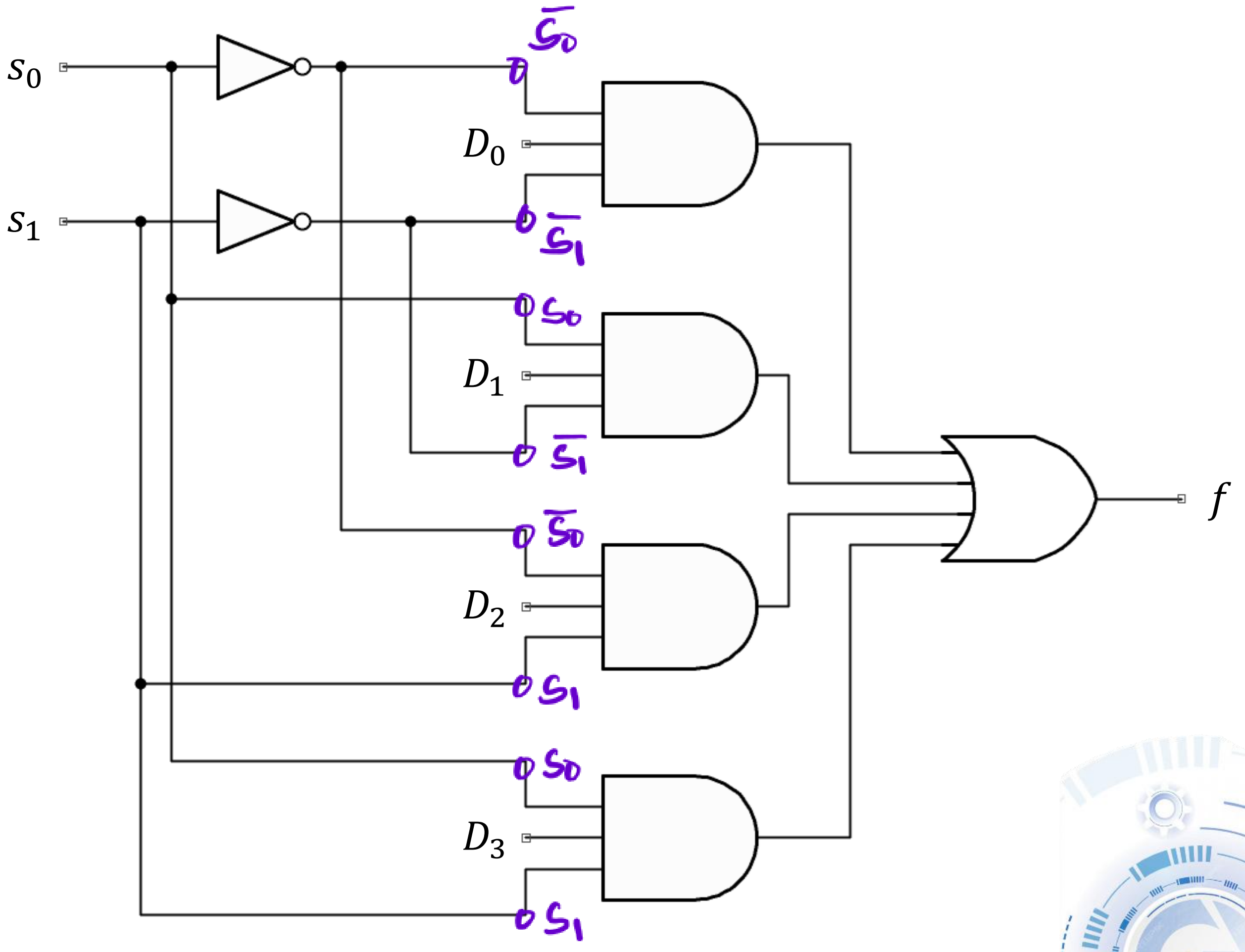
$\bar{S}_1 \bar{S}_0 D_0$

$\bar{S}_1 S_0 D_1$

$S_1 \bar{S}_0 D_2$

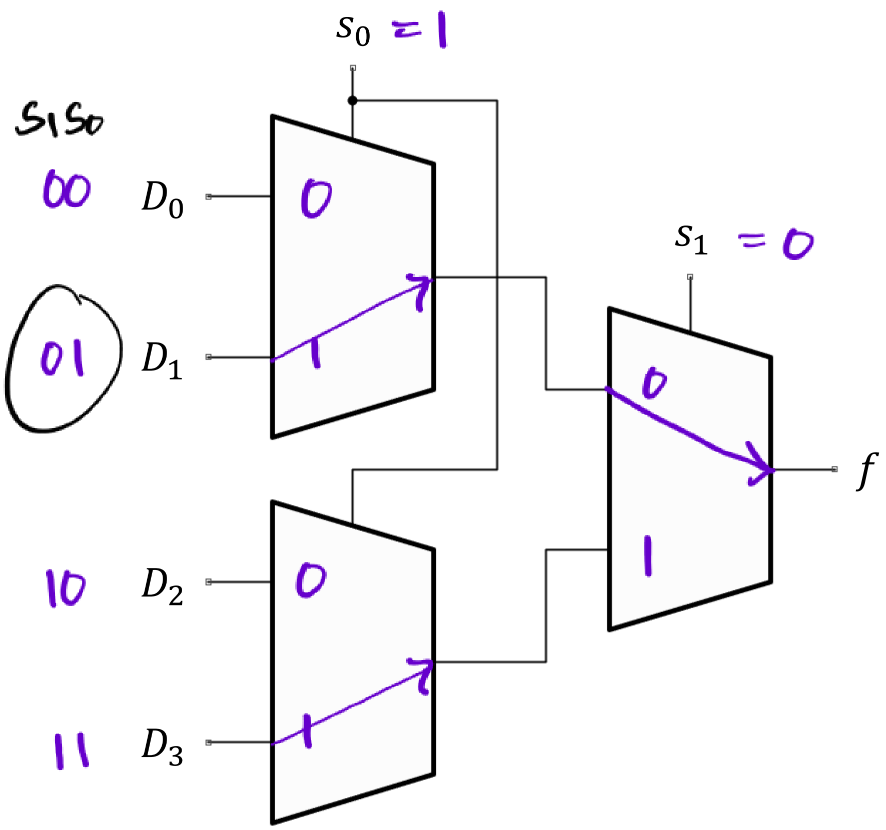
$S_1 S_0 D_3$

Sum-of-Products Implementation

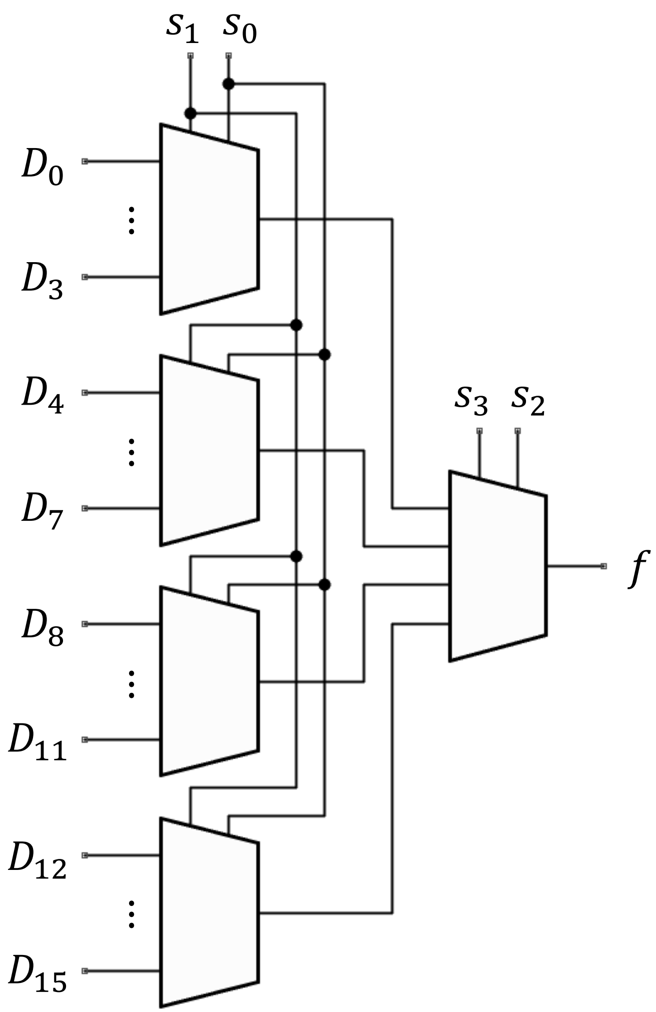


USING 2-TO-1 MUX

Using 2-to-1 MUX to build a 4-to-1 MUX



Using 2-to-1 MUX to build a 16-to-1 MUX

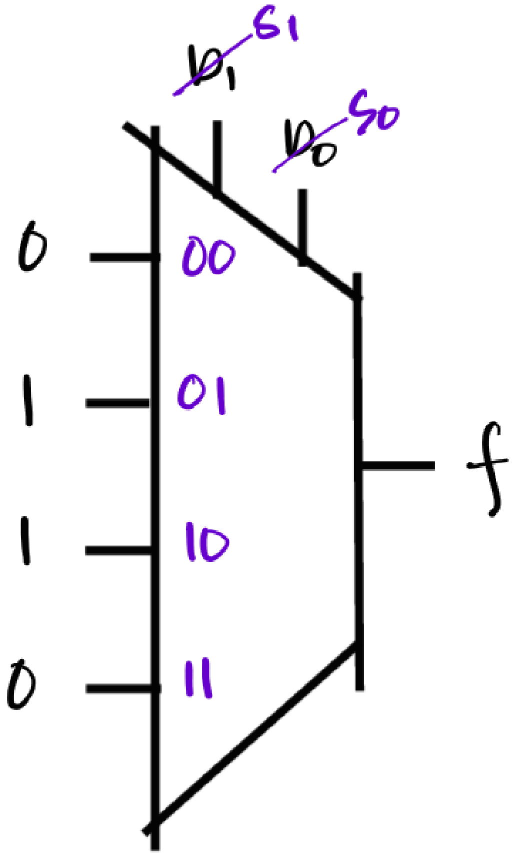


XOR LOGIC GATE

Implementation using 4-to-1 MUX

$\cancel{D_1}$ S_1	$\cancel{D_0}$ S_0	f
0	0 00	0
0	1 01	1
1	0 10	1
1	1 11	0

inputs



for 4 inputs

$\log_2 (n = 2^s)$

$s = \log_2 n$

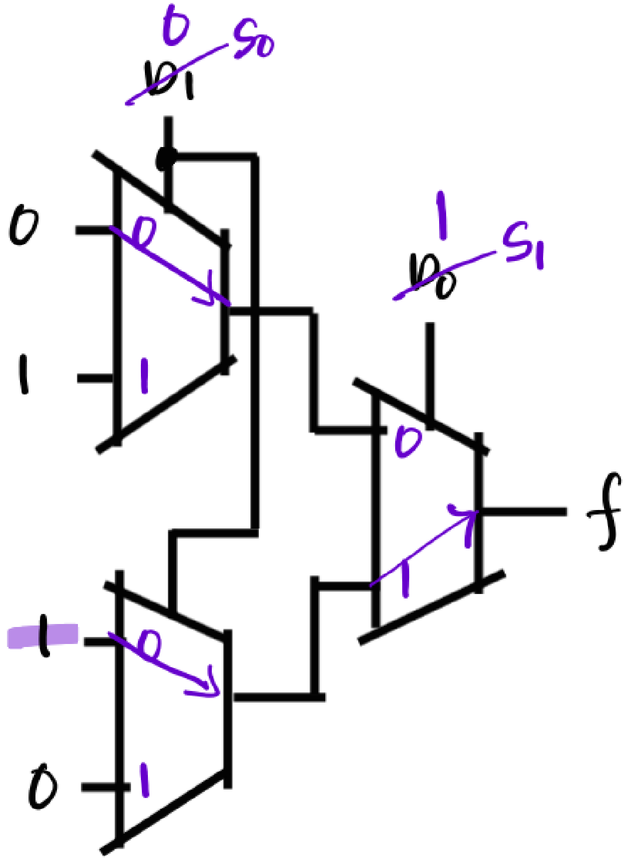
$s = \log_2 (4)$

$s = 2$

XOR LOGIC GATE

Implementation using 2-to-1 MUX

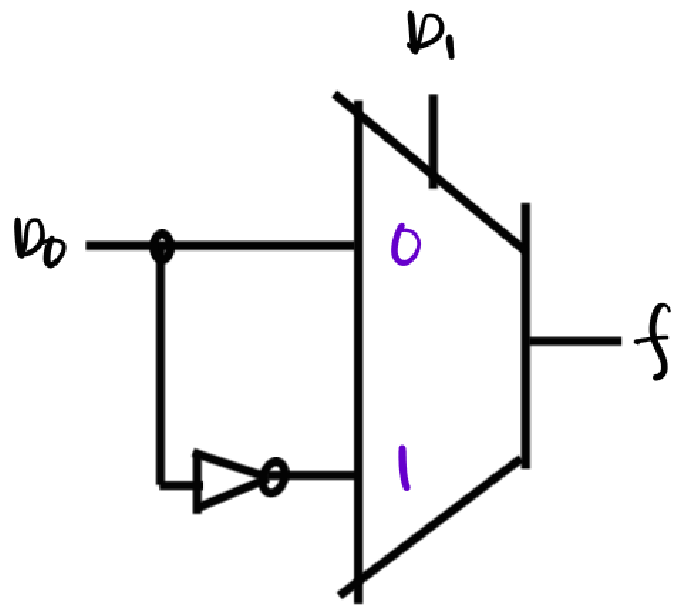
$\cancel{D_1}$ S_1	$\cancel{D_0}$ S_0	f
0	0	0
0	1	1
1	0	1
1	1	0



XOR LOGIC GATE

Implementation using 2-to-1 MUX

~~D_1~~ s	~~D_0~~ input	f
0	0 →	0
0	1 →	1
1	0 not	1
1	1 not	0

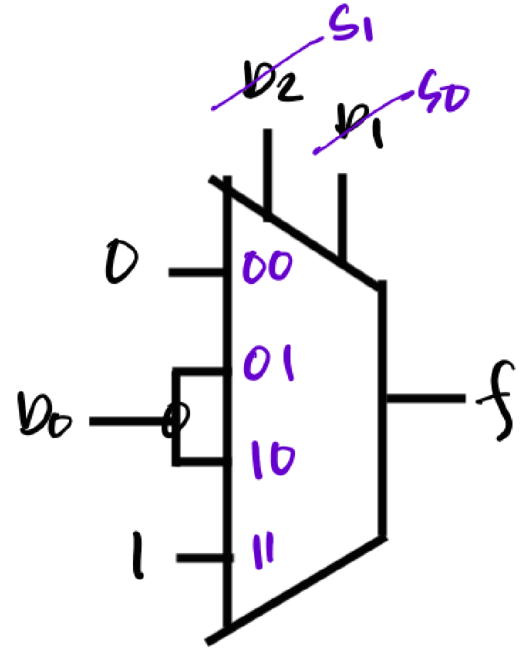


EXERCISE

Implement the logic function described by the truth table using a 4-to-1 multiplexer configuration.

~~D_2~~ S_1	~~D_1~~ S_0	~~D_0~~ input	f
0	0 } 00	0 x	0
0	0	1 x	0
0	1 } 01	0	0
0	1	1	1
1	0 } 10	0	0
1	0	1	1
1	1 } 11	0 x	1
1	1	1 x	1

Solution

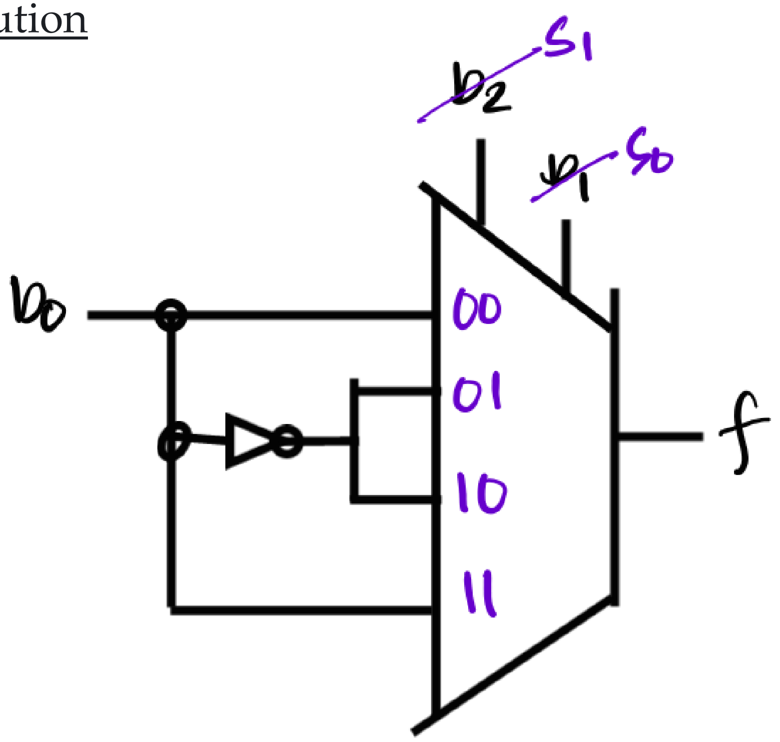


EXERCISE

Implement the three-input XOR described by the truth table using a 4-to-1 multiplexer configuration.

$\cancel{D_2}$ S_1	$\cancel{D_1}$ S_0	$\cancel{D_0}$ input	f
0	0 } 00	0	0
0	0	1	1
0	1 } 01	0 not	1
0	1	1 not	0
1	0 } 10	0 not	1
1	0	1 not	0
1	1 } 11	0	0
1	1	1	1

Solution



SHANNON'S EXPANSION THEOREM

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Shannon's expansion theorem states that any Boolean function $f(x_0, x_1 \dots, x_{n-1})$ can be written in the form:

$$\boldsymbol{f(x_0, x_1, \dots x_{n-1}) = \bar{x}_1 f_{\bar{x}_1} + x_1 f_{x_1}}$$

where

$f_{\bar{x}_1}$ = the cofactor of f with respect to $\bar{x}_1$

= $(x_0, \mathbf{0}, \dots x_{n-1})$

f_{x_1} = the cofactor of f with respect to x_1

= $(x_0, \mathbf{1}, \dots x_{n-1})$

selector

Expanding the given function with respect to x_1

$$f = x_0 x_1 + x_0 x_2 + x_1 x_2$$

$$f = x_1(x_0 \cdot 1 + x_0 x_2 + 1 \cdot x_2) + \bar{x}_1(x_0 \cdot 0 + x_0 x_2 + 0 \cdot x_2)$$

$A + AB = A$

$$f = x_1(\underline{x_0 + x_0 x_2} + x_2) + \bar{x}_1(x_0 x_2)$$

$$f = \overset{s}{\cancel{x_1}}(\underbrace{x_0 + x_2}_{\text{inputs}}) + \bar{x}_1(x_0 x_2)$$

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where

$f_{\bar{x}_1}$ = the cofactor of f with respect to $\bar{x}_1$

$= (x_0, \mathbf{0}, \dots x_{n-1})$

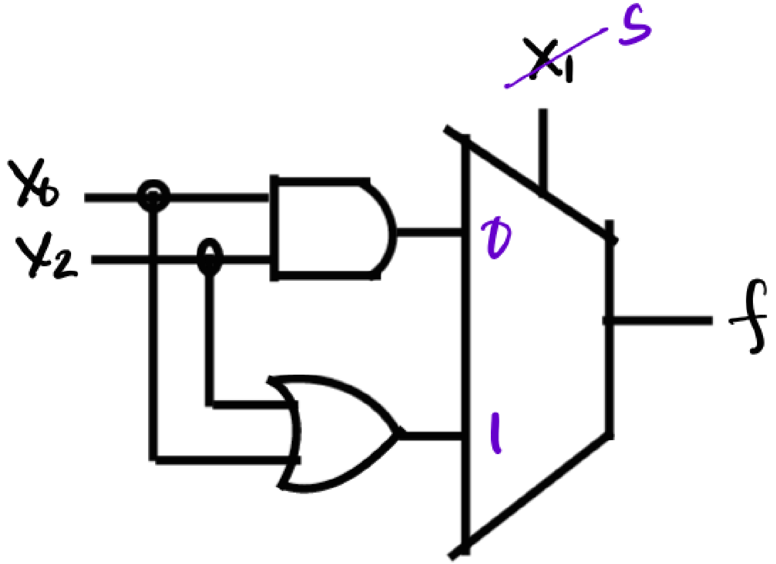
f_{x_1} = the cofactor of f with respect to x_1

$= (x_0, \mathbf{1}, \dots x_{n-1})$

selector

Expanding the given function with respect to x_1

$$f = \overset{s}{\cancel{x_1}}\underbrace{(x_0 + x_2)}_{\text{inputs}} + \bar{x}_1(x_0 x_2)$$



EXERCISE

Implement the logic function described by the truth table using Shannon's decomposition method.

s   inputs

D_2	D_1	D_0	f	Minterm
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{D}_2 D_1 D_0$
1	0	0	0	
1	0	1	1	$D_2 \bar{D}_1 D_0$
1	1	0	1	$D_2 D_1 \bar{D}_0$
1	1	1	1	$D_2 D_1 D_0$

Solution

$$f = \bar{D}_2 D_1 D_0 + D_2 \bar{D}_1 D_0 + D_2 D_1 \bar{D}_0 + D_2 D_1 D_0 \rightarrow 1$$

$$f = \bar{D}_2 D_1 D_0 + D_2 \bar{D}_1 D_0 + D_2 D_1$$

Decompose w/ D_2

$$f = D_2(\bar{D}_1 D_0 + D_1) + \bar{D}_2(D_1 D_0)$$

$$\bar{A}B + A = A + B$$

$$\boxed{f = D_2(D_0 + D_1) + \bar{D}_2(D_1 D_0)}$$

ans

EXERCISE

Implement the logic function described by the truth table using Shannon's decomposition method.

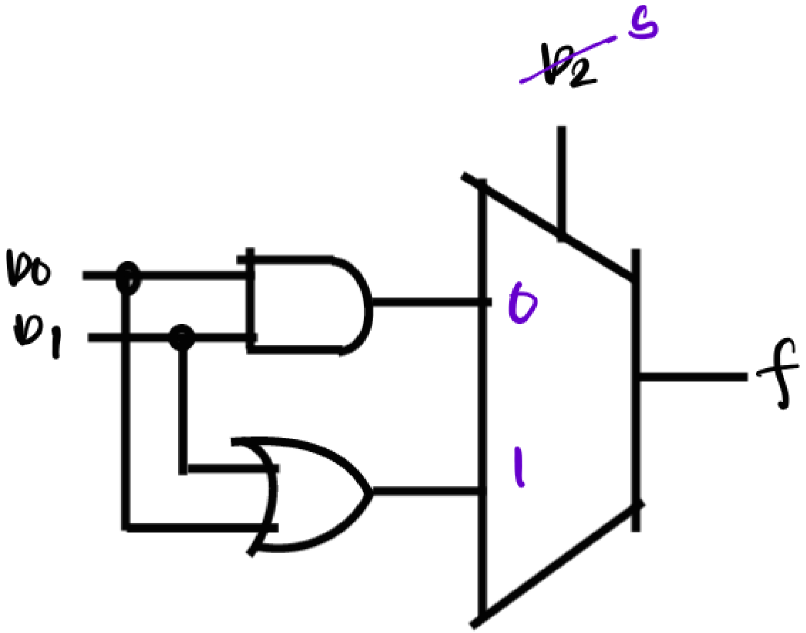
inputs

$\cancel{D_2}$ S	D_1	D_0	f	Minterm
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\overline{D_2} D_1 D_0$
1	0	0	0	
1	0	1	1	$D_2 \overline{D_1} D_0$
1	1	0	1	$D_2 D_1 \overline{D_0}$
1	1	1	1	$D_2 D_1 D_0$

Solution

$$\boxed{f = \overset{S}{\cancel{D_2}}(D_0 + D_1) + \overline{D_2}(D_1 D_0)}$$

ans



EXERCISE

Implement the logic function described by the truth table using Shannon's decomposition method.

$\cancel{D_2}$ S1	$\cancel{D_1}$ S0	$\cancel{D_0}$ input	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Solution

$$f = D_2(D_0 + D_1) + \bar{D}_2(D_1 D_0)$$

Decompose w/ D_1

$$f = D_2 D_1 (1) + \bar{D}_2 D_1 (D_0) +$$
$$D_2 \bar{D}_1 (D_0) + \bar{D}_2 \bar{D}_1 (0)$$

$$\boxed{f = \underbrace{D_2 D_1}_{11} + \underbrace{\bar{D}_2 D_1}_{01}(D_0) + \underbrace{D_2 \bar{D}_1}_{10}(D_0)}$$ ans

EXERCISE

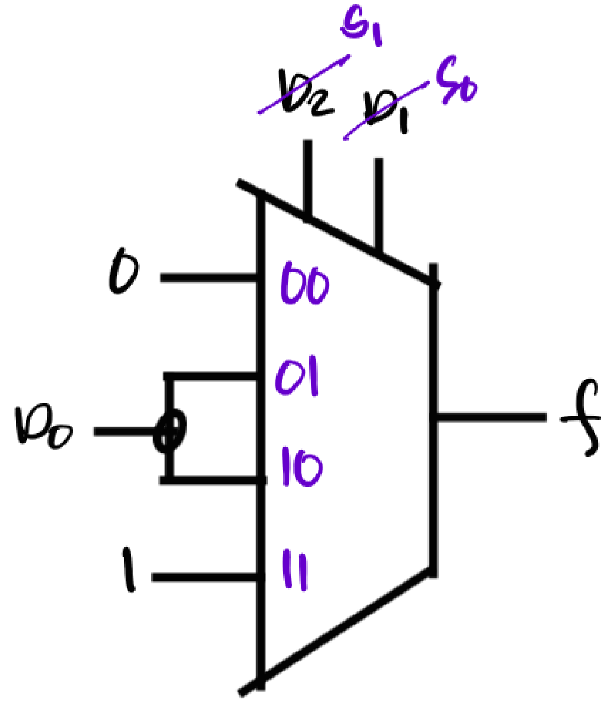
Implement the logic function described by the truth table using Shannon's decomposition method.

~~D_2~~ S1	~~D_1~~ S0	~~D_0~~ input	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Solution

$$\boxed{f = \underbrace{\overset{S_1}{\cancel{D_2}}\,\overset{S_0}{\cancel{D_1}}}_{1\,1} + \underbrace{\overline{D_2}\,D_1}_{0\,1}(D_0) + \underbrace{D_2\,\overline{D_1}}_{1\,0}(D_0)}$$

ans



EXERCISE

Implement the three-input XOR described by the truth table using a Shannon's decomposition method.

$\cancel{D_2}$ S1	$\cancel{D_1}$ S0	$\cancel{D_0}$ input	f	Minterm
0	0	0	0	
0	0	1	1	$\bar{D}_2 \bar{D}_1 D_0$
0	1	0	1	$\bar{D}_2 D_1 \bar{D}_0$
0	1	1	0	
1	0	0	1	$D_2 \bar{D}_1 \bar{D}_0$
1	0	1	0	
1	1	0	0	
1	1	1	1	$D_2 D_1 D_0$

Solution

Decompose w/ D_2

$$f = \bar{D}_2(\bar{D}_1 D_0 + D_1 \bar{D}_0) + D_2(\bar{D}_1 \bar{D}_0) + D_2(D_1 D_0)$$

Decompose w/ D_1

$$f = \bar{D}_2 \bar{D}_1 (D_0) + D_2 \bar{D}_1 (\bar{D}_0) + \bar{D}_2 D_1 (\bar{D}_0) + D_2 D_1 (D_0)$$

$$\boxed{f = \bar{D}_2 \bar{D}_1 (D_0) + D_2 \bar{D}_1 (\bar{D}_0) + \bar{D}_2 D_1 (\bar{D}_0) + D_2 D_1 (D_0)}$$

ans

EXERCISE

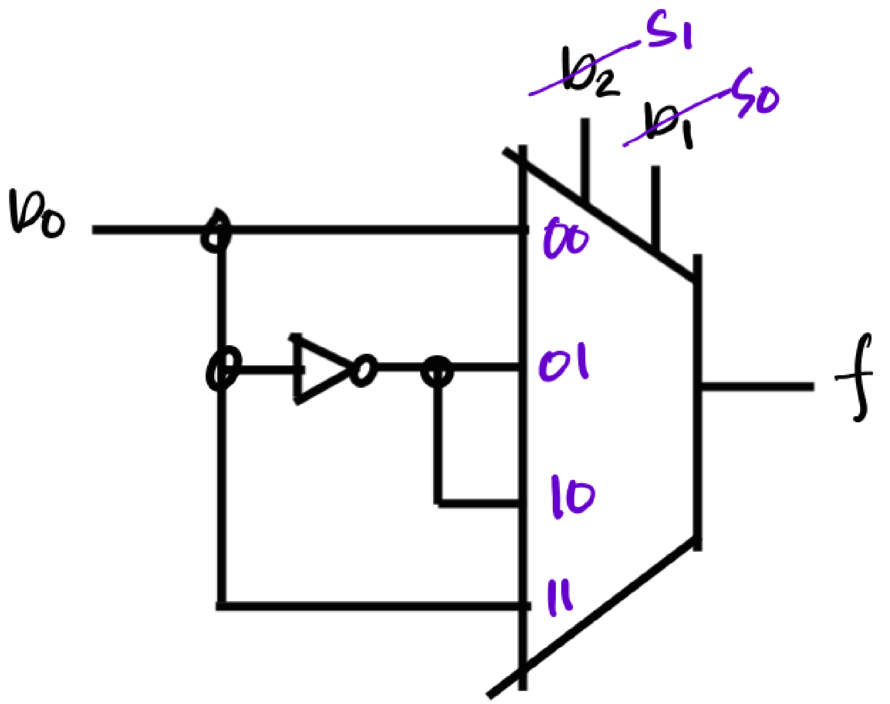
Implement the three-input XOR described by the truth table using a Shannon's decomposition method.

~~D_2~~ S_1	~~D_1~~ S_0	~~D_0~~ input	f	Minterm
0	0	0	0	
0	0	1	1	$\bar{D}_2\bar{D}_1 D_0$
0	1	0	1	$\bar{D}_2 D_1 \bar{D}_0$
0	1	1	0	
1	0	0	1	$D_2\bar{D}_1\bar{D}_0$
1	0	1	0	
1	1	0	0	
1	1	1	1	$D_2 D_1 D_0$

Solution

$$\boxed{f = \overset{00}{\bar{D}_2\bar{D}_1(D_0)} + \overset{10}{D_2\bar{D}_1(\bar{D}_0)} + \underset{01}{\bar{D}_2 D_1(\bar{D}_0)} + \underset{11}{D_2 D_1(D_0)}}$$

ans



EXERCISE

Design a full-adder circuit using Shannon's decomposition method and simulate it in Multisim using the 74151 multiplexer.

~~A~~ S_1	~~B~~ S_0	~~C_{in}~~ input	C_{out}	Σ	Minterm
0	0	0	0	0	
0	0	1	0	1	
0	1	0	0	1	
0	1	1	1	0	$\bar{A}BC_{in}$
1	0	0	0	1	
1	0	1	1	0	$A\bar{B}C_{in}$
1	1	0	1	0	$AB\bar{C}_{in}$
1	1	1	1	1	ABC_{in}

Solution

Mux for Cout

$$C_{out} = \bar{A}(BC_{in}) + A(\bar{B}C_{in} + B\bar{C}_{in} + BC_{in})$$

$$C_{out} = \bar{A}(BC_{in}) + A(B + \bar{B}C_{in})$$

$$X + \bar{X}Y = X + Y$$

$$C_{out} = \bar{A}(BC_{in}) + A(B + C_{in})$$

$$C_{out} = \bar{A}\bar{B}(0) + A\bar{B}(C_{in}) + \bar{A}B(C_{in}) + AB(1)$$

$$\boxed{C_{out} = A\bar{B}(C_{in}) + \bar{A}B(C_{in}) + AB}$$

ans

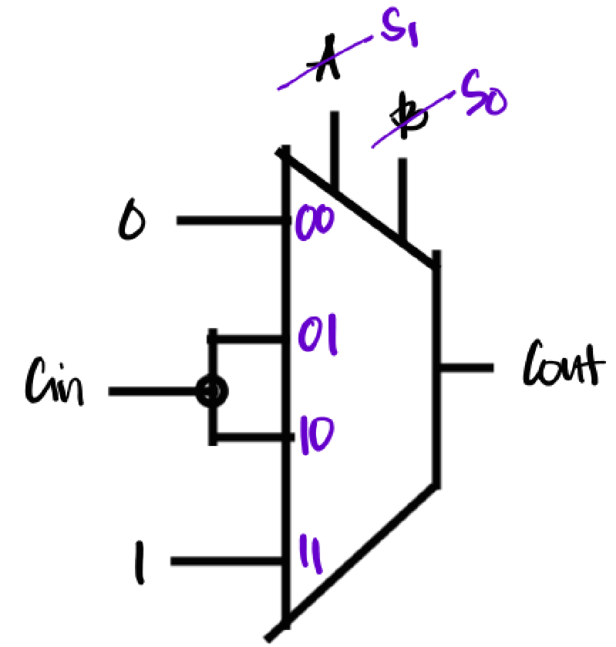
EXERCISE

Design a full-adder circuit using Shannon's decomposition method and simulate it in Multisim using the 74151 multiplexer.

$\cancel{A}$ S_1	$\cancel{B}$ S_0	$\cancel{C_{in}}$ input	C_{out}	Σ	Minterm
0	0	0	0	0	
0	0	1	0	1	
0	1	0	0	1	
0	1	1	1	0	$\bar{A}BC_{in}$
1	0	0	0	1	
1	0	1	1	0	$A\bar{B}C_{in}$
1	1	0	1	0	$AB\bar{C}_{in}$
1	1	1	1	1	ABC_{in}

Solution

$$\boxed{C_{out} = \underbrace{A\bar{B}}_{10}(C_{in}) + \underbrace{\bar{A}B}_{01}(C_{in}) + \underbrace{AB}_{11}}$$



EXERCISE

Design a full-adder circuit using Shannon's decomposition method and simulate it in Multisim using the 74151 multiplexer.

~~A~~ S_1	~~B~~ S_0	~~C_{in}~~ input	C_{out}	Σ	Minterm
0	0	0	0	0	
0	0	1	0	1	$\bar{A}\bar{B}C_{in}$
0	1	0	0	1	$\bar{A}B\bar{C}_{in}$
0	1	1	1	0	
1	0	0	0	1	$A\bar{B}\bar{C}_{in}$
1	0	1	1	0	
1	1	0	1	0	
1	1	1	1	1	ABC_{in}

Solution

Mux for sum

$$Sum = \bar{A}(\bar{B}C_{in} + B\bar{C}_{in}) + A(\bar{B}\bar{C}_{in} + BC_{in})$$

$$\boxed{Sum = \bar{A}\bar{B}(C_{in}) + A\bar{B}(\bar{C}_{in}) + \bar{A}B(\bar{C}_{in}) + AB(C_{in})}$$

ans

EXERCISE

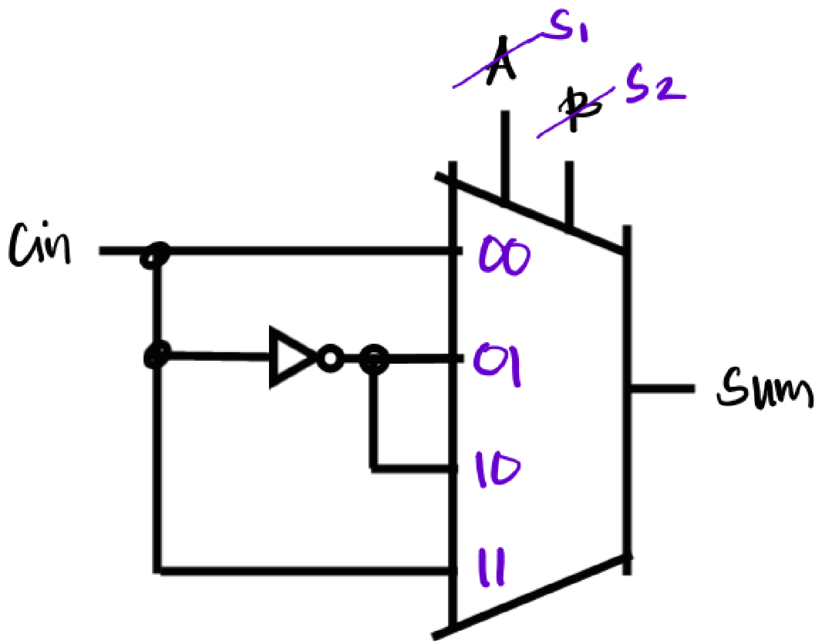
Design a full-adder circuit using Shannon's decomposition method and simulate it in Multisim using the 74151 multiplexer.

~~A~~ S_1	~~B~~ S_0	~~C_{in}~~ input	C_{out}	Σ	Minterm
0	0	0	0	0	
0	0	1	0	1	$\bar{A}\bar{B}C_{in}$
0	1	0	0	1	$\bar{A}B\bar{C}_{in}$
0	1	1	1	0	
1	0	0	0	1	$A\bar{B}\bar{C}_{in}$
1	0	1	1	0	
1	1	0	1	0	
1	1	1	1	1	ABC_{in}

Solution

$$\boxed{Sum = \underbrace{\bar{A}\bar{B}(C_{in})}^{00} + \underbrace{A\bar{B}(\bar{C}_{in})}^{10} + \underbrace{\bar{A}B(\bar{C}_{in})}_{01} + \underbrace{AB(C_{in})}_{11}}$$

ans



LABORATORY